

Hands-01 Git Hands-On Lab

Aim

The aim of this lab is to learn the basic concepts of Git, configure Git on a local machine, create a local repository, track files, commit changes, and connect the local repository to a remote GitHub repository. This lab also helps understand how version control is used to manage source code efficiently.

Objectives

- Install and verify Git.
- Configure Git with username and email.
- Learn basic Git commands.
- Create a local Git repository.
- Create and track files using Git.
- Commit changes to the local repository.
- Connect the local repository to GitHub.
- Push the project from the local repository to the remote repository.

Prerequisites

- Git installed on the system.
- GitHub account.
- Internet connection.
- Command Prompt or Git Bash.

Git Commands Used

`git --version`

Checks whether Git is installed and displays the installed version.

`git config --global user.name "Adharshinivankayala"`

Configures the Git username.

`git config --global user.email "204443666+Adharshinivankayala@users.noreply.github.com"`

Configures the Git email.

`git config --global --list`

Displays the current Git configuration.

Creating a Local Repository

Navigate to the Desktop.

`cd Desktop`

Create a new project folder.

```
mkdir GitDemo
```

Move into the project folder.

```
cd GitDemo
```

Initialize the Git repository.

```
git init
```

Git creates a hidden .git folder that stores all repository information.

Creating a File

Create a text file.

```
echo "Welcome to Git" > welcome.txt
```

Verify that the file is created.

```
dir
```

Display the contents of the file.

```
type welcome.txt
```

Checking Repository Status

```
git status
```

This command displays the current status of the repository, including tracked and untracked files.

Initially, welcome.txt appeared as an untracked file.

Adding the File to the Staging Area

```
git add welcome.txt
```

This command moves the file from the working directory to the staging area so that Git can track it.

Creating the First Commit

```
git commit -m "Added welcome.txt"
```

This command creates the first snapshot of the repository by saving the staged changes with a commit message.

Creating a Remote Repository

A new repository named **GitDemo** was created on GitHub without adding a README file.

Connecting Local Repository to GitHub

```
git remote add origin https://github.com/Adharshinivankayala/GitDemo.git
```

This command links the local repository with the remote GitHub repository.

Renaming the Default Branch

```
git branch -M main
```

This command renames the default branch from master to main.

Updating Git Email

Since GitHub rejected the push due to email privacy settings, the Git email was updated.

```
git config --global user.email "204443666+Adharshinivankayala@users.noreply.github.com"
```

Verify the configuration.

```
git config --global --list
```

Updating the Commit Author

```
git commit --amend --reset-author --no-edit
```

This command updates the author information of the previous commit without changing the commit message.

Pushing the Repository

Initially, the push failed because the commit history had changed after amending the commit.

The repository was successfully pushed using:

```
git push -u origin main --force
```

The repository was uploaded successfully to GitHub.

Verifying the Repository

```
git status
```

Output:

On branch main

nothing to commit, working tree clean

This confirms that all changes have been committed and the local repository is synchronized.

Commands Executed During the Lab

```
git --version
```

```
git config --global user.name "Adharshinivankayala"
```

```
git config --global user.email "204443666+Adharshinivankayala@users.noreply.github.com"
```

```
git config --global --list
```

```
cd Desktop
```

```
mkdir GitDemo
```

```
cd GitDemo
```

```
git init
```

```
echo "Welcome to Git" > welcome.txt
```

```
dir
```

```
type welcome.txt
```

```
git status
```

```
git add welcome.txt
```

```
git commit -m "Added welcome.txt"
```

```
git remote add origin https://github.com/Adharshinivankayala/GitDemo.git
```

```
git branch -M main
```

```
git commit --amend --reset-author --no-edit
```

```
git push -u origin main --force
```

```
git status
```

What Was Implemented

- Installed and verified Git on the system.
- Configured Git username and email.
- Created a local Git repository.
- Created a sample file named welcome.txt.
- Added the file to the staging area.
- Committed the file to the local repository.
- Created a remote repository on GitHub.
- Connected the local repository to the remote repository.
- Renamed the default branch to main.
- Updated the commit author using the GitHub no-reply email.
- Successfully pushed the project to GitHub.
- Verified that the local repository and remote repository were synchronized.

Result

The Git Hands-On Lab was successfully completed. A local Git repository was created, configured, connected to GitHub, and synchronized with a remote repository. The project containing welcome.txt was successfully committed and pushed to GitHub, demonstrating the complete Git workflow from repository initialization to remote repository integration.

```
Command Prompt
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Writing objects: 100% (3/3), 259 bytes | 259.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/Adharshinivankayala/GitDemo.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.

C:\Users\vanka\Desktop\GitDemo>git config --global --list
user.name=Adharshinivankayala
user.email=204443666+Adharshinivankayala@users.noreply.github.com
filter.lfs.required=true
filter.lfs.clean=git-lfs clean -- %f
filter.lfs.smudge=git-lfs smudge -- %f
filter.lfs.process=git-lfs filter-process

C:\Users\vanka\Desktop\GitDemo>git commit --amend --reset-author --no-edit
[main f5cd28d] Added welcome.txt
1 file changed, 1 insertion(+)
create mode 100644 welcome.txt

C:\Users\vanka\Desktop\GitDemo>git push -u origin main
To https://github.com/Adharshinivankayala/GitDemo.git
 ! [rejected]        main -> main (non-fast-forward)
error: failed to push some refs to 'https://github.com/Adharshinivankayala/GitDemo.git'
hint: Updates were rejected because the tip of your current branch is behind
hint: its remote counterpart. If you want to integrate the remote changes,
hint: use 'git pull' before pushing again.
hint: See the 'Note about fast-forwards' in 'git push --help' for details.

C:\Users\vanka\Desktop\GitDemo>git push -u origin main --force
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Writing objects: 100% (3/3), 258 bytes | 258.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/Adharshinivankayala/GitDemo.git
 + 5389ad7...f5cd28d main -> main (forced update)
branch 'main' set up to track 'origin/main'.

C:\Users\vanka\Desktop\GitDemo>
```